



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X
SUBJECT: BIOLOGY

FULL MARKS: 40
DATE: 29.4.24

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

1. Choose the correct answer: [5]

- a. Assertion (A) - G₁ phase is the shortest phase of Interphase.
Reason (R) - G₁ phase is the phase where synthesis of RNA and protein take place and cytoplasm increases in volume.
- | | |
|----------------------------|------------------------------|
| i. Both A and R are false | ii. A is true and R is false |
| iii. Both A and R are true | iv. A is false and R is true |
- b. The condition in which excess glucose passes through urine:
- | | |
|----------------|-----------------|
| i. Albuminuria | ii. Diuresis |
| iii. Urochrome | iv. Glycosuria. |
- c. Sheena was going through the cell division pattern of a cell. She observed that the cells are dividing in a particular pattern. How many mitotic divisions are required for 1 skin cell to form 16 cells?
- | | |
|--------|--------|
| i. 2 | ii. 4 |
| iii. 6 | iv. 8. |
- d. Assertion (A) – Cortex region of the kidney is dotted.
Reason (R) – Cortex region of the kidney is dotted because of the presence of Loop of Henle and Bowman's Capsule.
- | | |
|----------------------------|------------------------------|
| ii. Both A and R are false | ii. A is true and R is false |
| iii. Both A and R are true | iv. A is false and R is true |
- e. Hormone which regulates carbohydrate, protein and fat metabolism:
- | | |
|---------------------|------------------------|
| i. Glucocorticoids | ii. Mineralocorticoids |
| iii. Sex corticoids | iv. Adrenaline |

2. Choose the odd one out and write the correct category to which the others belong:

- a. FSH, TSH, ACTH, Oxytocin.
b. Urea, Carbonic acid, Uric acid, Creatinine.
c. Loop of Henle, DCT, PCT, Renal Artery.

[5]

- d. Alpha cells, Beta cells, Delta cells, Glucagon.
- e. DNA, Nucleosome, Histone, Centrosome.

3. Fill in the blanks: [5]

One vertical half of the duplicated chromosome is called ____a____. These two vertical halves are joined by a ____b____. The components of chromatin are ____c____ and ____d____. The scientist who discovered chromosome is ____e____.

4. Match the correct terms of Column I with Column II. Please rewrite the correct pairs: [5]

Column I	Column II
i. Vasa recta	Deoxyribonucleic acid
ii. DNA	Main nitrogenous waste product
iii. Oxytocin	Nitrogenous base, Pentose sugar, Phosphate
iv. Urea	Contraction of uterine wall
v. Nucleotide	Secondary capillaries around loop of Henle
	Hormone from anterior pituitary
	Glucose in urine

5. Answer the following questions:

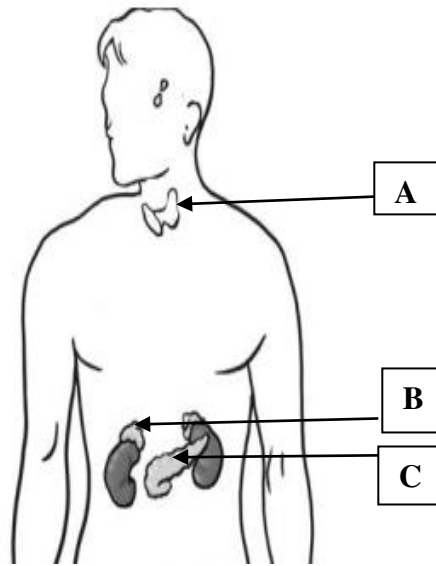
- a. Define Genes. [1]
- b. Differentiate between Prophase and Telophase on the basis of nuclear membrane. [2]
- c. Give reasons: [2]
During summer urine is thicker than in winter.
- d. Mention the two nitrogenous bases which have double hydrogen bonds between them. [2]
- e. Draw a neat and well labelled diagram of double helical structure of DNA. [3]

6. Answer the following questions:

- a. Write the full form of LH. [1]
- b. Mention two important features of Anaphase. [2]
- c. Mention two symptoms of the condition Myxoedema. [2]
- d. Name the gland from which we get adrenaline. Why is adrenaline known as an emergency hormone? [2]

e. Answer the following questions by observing the diagram.

[3]



- i. Identify the glands 'A'.**
- ii. Mention the location of 'B' and 'C'.**